

Room Backgrounds — Wiring into the Game (Cursor)

RoboPet · Visual Pass · Drop the three Pixellab rooms in as backdrops behind the wandering dog.

1. Which file is which

- Couch + holographic TV + lamp + plant -> Living Room -> living_room.png
- Monitors + beakers on shelves -> Laboratory -> laboratory.png
- 3D-printer + amber robotic arms -> Fabricator -> fabricator.png

2. Place assets and declare them

Put the PNGs in assets/images/rooms/:

```
assets/images/rooms/living_room.png laboratory.png fabricator.png
```

Add the folder to pubspec (keep your dog folders too), then run flutter pub get:

```
flutter:  
  assets:  
    - assets/images/rooms/
```

3. lib/game/rooms/living_room.dart (updated)

Replace the dark background rectangle with the room sprite, and remove the old floor line (the art has its own floor):

```
import 'package:flame/components.dart';  
import 'package:flame/game.dart';  
  
import '../components/robot_dog.dart';  
  
class LivingRoomWorld extends World {  
  @override  
  Future<void> onLoad() async {  
    await super.onLoad();  
  
    // Room backdrop: fills the same 400 x 700 box as the old dark rectangle.  
    add(SpriteComponent(  
      sprite: await Sprite.load('rooms/living_room.png'),  
      size: Vector2(400, 700),  
      position: Vector2(-200, -350),  
      priority: -10,  
    ));  
  
    // Robot dog standing on the painted floor.  
    // Nudge the y value until the feet sit on the floor (try 140 to 160).  
    add(RobotDog(position: Vector2(0, 150)));  
  }  
}
```

4. Seat the dog on the painted floor

The room art puts the floor in the lower third, so the dog needs to move down from its old spot. Start at $y = 150$ and nudge until the feet rest on the floor (try 140 to 160). The dog still wanders left and right on that same line.

5. Apply to Laboratory and Fabricator

Do the exact same swap in your Laboratory and Fabricator worlds, loading `rooms/laboratory.png` and `rooms/fabricator.png`. Their floors sit a little lower, so try a dog y around 170 to 200 and tune.

6. Optional - per-room wander range

If the dog walks past the edges of the painted floor, limit how far it roams. Make `_wanderRange` a constructor value in `robot_dog.dart`:

```
// 1) Change the field from a const into a final set by the constructor:
//   DELETE: static const double _wanderRange = 130.0;
//   ADD:    final double _wanderRange;

// 2) Update the constructor to accept it:
RobotDog({required Vector2 position, double wanderRange = 130})
  : _wanderRange = wanderRange,
    super(
      position: position,
      size: Vector2.all(80),
      anchor: Anchor.bottomCenter,
      priority: 10,
    );

// Then each room can pass: RobotDog(position: ..., wanderRange: 110)
```

Then pass `RobotDog(position: Vector2(0, 150), wanderRange: 110)` per room.

7. Cursor task

Paste this to Cursor and attach this PDF:

```
Add room backgrounds to my Flutter + Flame game.
1. I added assets/images/rooms/living_room.png, laboratory.png, fabricator.png.
   Register assets/images/rooms/ in pubspec.yaml and run flutter pub get.
2. In lib/game/rooms/living_room.dart, replace the dark background RectangleComponent
   with a SpriteComponent that loads rooms/living_room.png, size Vector2(400, 700),
   position Vector2(-200, -350), priority -10 so it sits behind the dog. Remove the old
   floor line RectangleComponent.
3. Move the RobotDog down so it stands on the painted floor (start position Vector2(0,
   150); I will fine-tune).
4. Apply the same backdrop pattern to my Laboratory and Fabricator worlds using
   rooms/laboratory.png and rooms/fabricator.png, with the dog y around 170 to 200.
5. Hot-restart so the new assets load.
```